MATRIX-RING BLUEPRINT AUDIT

Independent mathematical, structural, theorem, and prior-art review

Audited proposal	Constacyclic codes over M (F [u]/(u)) and quantum codes over F
Audit date	12 November 2026
Overall decision	**REJECT AS SUBMITTED DO NOT PROCEED IN ITS CURRENT FORM**

Scope: audit only. No construction, proof development, code-parameter search, or modification of the uploaded blueprint was undertaken. The original uploaded PDF remains unchanged.

Decision in brief: the specified matrix ring is noncommutative and nonlocal; the blueprint treats it as a commutative local ring with residue field F . That foundational mismatch invalidates the proposed component, ideal, duality, Gray-map, and quantum-code chain as submitted. Novelty is not established; a revised direction requires researcher approval.

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Literature review is targeted, not exhaustive. Source-extraction and theorem-verification limits are stated in the reports.

Complete blueprint audit

Audited document: Blueprint A: Constacyclic codes over M (F [u]/ u^k) and quantum codes over F , 7 pages, dated 12 November 2026.
Audit date: 12 November 2026.
Scope: independent review of the supplied blueprint and targeted source/literature verification. This is an audit only. No construction, proof development, parameter search, code search, or change to the original blueprint was undertaken. The uploaded original PDF was not modified.

Final decision

REJECT AS SUBMITTED do not proceed with this blueprint in its current form. The proposed ring model is incompatible with the stated matrix ring, and that error propagates into the proposed constacyclic decomposition, ideal classification, duality, Gray-map and quantum-code claims. The research direction is not declared impossible; it would require a new algebraic formulation and a fresh novelty review after researcher approval.

Executive findings

1 **Critical algebra error.** For A = F [u]/(u^k), the ring M (A) is noncommutative and is not local. Its Jacobson radical is M (uA) (zero for k=1); its quotient is M (F), not F . Its center is the scalar matrices A I . The claimed commutative local presentation F [u,v]/(u^k,v) therefore cannot be a ring presentation of the stated matrix ring.

2 **The proposed order-5 multipliers conflict with centrality.** The four nonidentity fifth roots in an embedded F are units, but that embedded field is not central in the matrix ring. Central units are scalar matrices from A ^ , whose order is a power of 2; none has order 5. A noncentral shift can be defined with a side convention, but the blueprint's ordinary polynomial-quotient/ideal and CRT route does not follow from it.

3 **The structure/cardinality plan is conditional on the wrong ring.** The proposed factors K[u,v]/(u^k,v), local ideals, monomial counts, and Hensel/CRT argument cannot be used for M (A) without a separate, correctly sided module-theoretic derivation.

4 **Duality and quantum transfer are not established.** The field-level identity ^{-1}= for =1 is correct when = in F . It does not by itself define an involution or Hermitian form on the noncommutative matrix ring, prove a dual formula, or show a Gray map preserves orthogonality.

5 **Novelty is not established.** Direct prior work exists for the k=2 ring (2022) and a 2025 ISIT paper covers the k=3 cyclic-code case. A 2026 arXiv preprint treats cyclic codes for general k, and a 2026 Quantum Information Processing article develops quantum codes from cyclic codes over M -rings. These do not, by themselves, settle the novelty of a valid M constacyclic/quantum extension; the claimed gap is nevertheless too narrow and insufficiently checked.

Section-by-section audit

Page 1 overview, abstract, preliminaries, choice of

The cardinality identity |M (A)|=2^(16k)=16^(4k) is correct. It permits a **set-size** count of 4k field symbols per ring entry. It does not imply a local-ring isomorphism, a canonical F -vector-space structure, or any Gray-map property.

Local with residue field F and maximal ideal u,v are false for the stated matrix ring. See the algebra audit.

A unit exactly when its image in F is nonzero is false for M (A). A matrix is a unit exactly when its reduction modulo u is invertible in M (F).

The four nontrivial order-5 elements exist in F ^{ }, but they cannot be treated as central scalars in M (A). The draft abstract's phrase units with =1 in the residue field F uses a residue field the matrix ring does not have.

The opening claim that constacyclic codes give quantum codes that cyclic codes cannot reach is a hypothesis, not an established consequence. No construction or parameter comparison supports it.

Pages 2 3 conjugation, factorization, code structure, cardinality, duals

The source's F -coordinate normal form must not be read as a commutative truncated polynomial ring. The 2026 preprint describes the cyclic-algebra relation ae=e (a), with e =1; its nilpotent coordinate v=1+e is not a central indeterminate. The distinction is foundational.

The claim that the extension a a , fixing u,v, is a ring automorphism only if F commutes with v is not a valid general criterion. In a skew presentation one must check the defining relations. More importantly, a Hermitian form on a noncommutative module requires a specified involution/anti-involution, side convention, and scalar target; none is supplied.

For a commutative field polynomial in characteristic two, odd n and nonzero do make x - separable. The Hensel/CRT argument in the blueprint, however, is applied to an asserted commutative local ring and cannot be transferred to the actual matrix ring by cardinality or by a module normal form.

The displayed S =K[u,v]/ u^k,v decomposition, the classification of its ideals, and the product formula for |C| are therefore **not verified theorems about the stated R**. The monomial-ideal count is valid only inside the proposed commutative quotient after that quotient has been justified; it does not classify all ideals even there.

The Euclidean/Hermitian constacyclic-dual rules and component pairing are conditional on the exact shift, module side, and form. The proposed annihilator criterion also leaves left/right annihilators and the component conjugation map unspecified.

Page 4 Gray map and quantum construction

N'=4kn is the correct symbol count **if** there is a bijection R F ^{4k}. It does not prove additivity, F -linearity, Hamming isometry, a quasi-twisted index, or orthogonality preservation.

The proposed equalities (C^(perp_H))= (C)^{perp_H} and (C) being F -linear are central unproved obligations, not routine properties of a Gray map. The attached 2026 matrix-ring quantum paper uses the reversed-index product a,brangle_R=sum a b ; its sampled Theorems 12 15 state a dual-size formula, a dual-containing criterion, distance-preserving maps for its k=3,2 cases, and a dual-transfer identity. The extracted Theorem 15 proof is incomplete mid-argument, so those are recorded as source claims, not independently certified here. Its different product/map pair is a relevant warning, not a proof for M .

The standard Hermitian construction is correctly stated conditionally: an [N,K,d]_{16} linear code D with D^(perp_H) D gives a 4-ary quantum code of dimension 2K-N. The quantum distance is the minimum weight of a nontrivial logical coset (often Dsetminus D^(perp_H)); it is at least d_H(D), but exact distance and equality/purity must be established.

The quantum Singleton bound and defect formula are standard once the parameters are valid. They do not establish that any such code exists.

Pages 5 6 parameter claims, search plan, outline, risks

The image length N=4kn does not depend on . A different multiplier may change algebraic structure or available component choices in a valid setting, but new lengths reachable only for ne1 is unsupported as written.

The proposed search grid is arithmetically consistent with Nle400 at k=4,n=25, but its parameter assumptions depend on the invalid order-5 central family and unproved component model.

Enumeration of monomial ideals would explore only a restricted subfamily. It cannot certify a classification or a necessary-and-sufficient dual-containment theorem for all codes. A finite-field distance computation also cannot validate a flawed algebraic encoding or map.

Grassl-table comparisons need an explicit convention for exact distance versus lower bounds, domination, field size, and the comparison snapshot/date. No parameter search was performed, as required by the audit-only constraint.

The work order correctly recognizes the need to inspect maps and full papers before research execution, but the proposed sequence starts from a false ring model. The fallback to monomial ideals does not repair that foundation.

Page 7 journals and Q1 readiness

Journal names and quartiles are targets to check, not evidence of readiness. Quartile depends on database, subject category, and year. The current plan has no valid main theorem, verified orthogonality transfer, complete novelty analysis, or validated examples; it is not Q1-ready.

Direct answers

Reliable? No. The ring/local/residue-field/unit claims are contradicted by the actual matrix-ring structure, and the later results rely on them.

Novel? Not established. There is substantial direct prior art for cyclic codes over the same ring family, plus closely related quantum constructions. The exact novelty of a correctly reformulated constacyclic/quantum contribution remains unverified; absence of a hit in a targeted search is not proof of novelty.

Ready for research execution or Q1 submission? No. The algebra, shift convention, involution/inner product, component decomposition, Gray map, and literature positioning must be rebuilt and approved first.

Audit limits and principal references

The audit reviewed the full 7-page blueprint; the 30-page arXiv v1 preprint and its odd-length Euclidean-dual theorem; the attached 30-page 2026 QIP paper's ring/dual/quantum-map sections; publisher metadata and abstract for the 2022 M paper; its published correction; and targeted metadata/abstracts for related 2021 2026 work. The full body of the 2022 subscription article was not available for a line-by-line theorem audit. The extracted QIP Theorem 15 proof ends mid-argument and was not independently verified against rendered pages. The literature search was targeted, not exhaustive. No code-parameter or computational search was run.

Joshi, Patel & Prakash, arXiv:2602.18255v1, submitted 20 Feb 2026.

Patel, Prakash & Islam, Cyclic codes over M (F +u F) , Cryptography and Communications 14 (2022); published correction.

Joshi, Patel & Prakash, Quantum codes construction from cyclic codes over matrix rings , Quantum Information Processing 25, 207 (2026).

Joshi, Patel & Prakash, Construction of new codes over F from cyclic codes over M (F +u F +u F), IEEE ISIT 2025; also cited as Ref. 19 in the 2026 QIP paper.

Mathematical correctness audit

Verdict: the blueprint contains a correct cardinality calculation and several standard field-level formulas, but its central ring model and the transfer of those formulas to the matrix ring are incorrect or unproved. This is a mathematical audit, not a repair or proof-development exercise.

Claim-by-claim review

1. Cardinality and symbol count correct, narrowly

For A = F [u]/ u^k , |A |=2^k. Therefore

|M (A)|=|A |^{16}=2^(16k)=16^(4k).

A bijection from one ring symbol to 4k symbols of F would be cardinality-compatible, and a length-n word would map to 4kn symbols. This establishes neither a ring isomorphism nor a linear, isometric, or orthogonality-preserving map.

2. Local-ring and residue-field claims false

The blueprint treats M (A) as a local commutative ring with residue field F . In fact, its Jacobson radical is M (uA) (zero for k=1) and its quotient is M (F). This quotient is a noncommutative matrix ring, not a field. The center of M (A) is A I . Since the claimed residue field and maximal ideal are wrong, the blueprint's unit criterion, Hensel/CRT components, and local-ideal model do not apply as stated.

3. Unit test false as stated

A matrix in M (A) is invertible exactly when its reduction modulo u is invertible in M (F). It is not enough that a purported F -component be nonzero. The latter test is for a commutative local ring with residue field F , which is not this ring.

4. Order-5 multiplier family field fact true, ring use invalid

The multiplicative group F ^{ } has order 15, so it has exactly four nonidentity elements of order 5. That field fact is correct. But the embedded F in M (F) is not central. The central units of M (A) are scalar matrices from A ^{ }; this group has order 1 for k=1 and 2^(k-1) for k>1, so it contains no element of order 5. Thus the proposed order-5 family cannot be used as central scalars in the blueprint's ordinary quotient/ideal framework.

A noncentral multiplier may still define a right-linear shift if the code is a right R-module and the multiplier acts on the left. That possibility requires an explicit definition and a separate module treatment; it does not validate R[x]/ x - as an ordinary quotient ring or the proposed CRT argument.

5. Separability and Hensel lifting conditional only

In characteristic two, for odd n and nonzero field , the derivative of x - is x , so the polynomial is separable over a field. This is a valid field-level observation. Hensel lifting as used in the blueprint additionally requires a valid commutative coefficient ring (or a rigorously specified noncommutative analogue), a central polynomial/quotient setup, and compatible factorization. The blueprint has not supplied those prerequisites for M (A).

6. Component rings and ideal counts conditional on a false premise

The ring K[u,v]/ u^k,v is a finite commutative local complete-intersection/Frobenius ring. For a monomial ideal in that quotient, its additive size can be counted as |K|^{m }, where m counts the monomial basis elements in the ideal. Those statements concern the proposed commutative quotient, not automatically the matrix-ring code components. The blueprint has not established the asserted component-ring isomorphism. Its monomial count also covers only monomial ideals; it is not a classification of all ideals or codes.

7. Dual of a constacyclic code correct in the usual field setting, not transferred

For a field code with a specified conventional -shift and standard Hermitian conjugation z z on F , the corresponding Hermitian dual multiplier is ^{-1}= . If =1, then = . This explains the proposed field-level motivation.

It does not establish the same formula for a noncommutative matrix-ring right module. The blueprint must first define the shift side, inner product, coefficient involution (or anti-involution), and any induced operation on . An entrywise field Frobenius on an embedded subfield is not, by itself, a specified involution on the full matrix ring.

8. Component pairing and annihilator criterion not verified

The root rule j -4j is a plausible reciprocal-conjugate rule in a particular field factorization, but the blueprint gives no root parametrization, lift, or proof that it describes the proposed components. Ann also needs a left/right convention for a noncommutative ring. The displayed equivalence between dual containment and component annihilator inclusion is a target, not a verified theorem.

9. Gray-map claims unproved

The statements (C) is F -linear, index 4k quasi-twisted, (C^(perp_H))= (C)^{perp_H}, and Hamming-distance preservation are independent obligations. A cardinality-compatible encoding or a Gray isometry for some weight does not imply any of them. The 2026 matrix-ring quantum paper (for M) explicitly distinguishes a first Gray map that fails to transfer duals from a different map/product pair designed to do so; this illustrates the issue.

10. Quantum parameters standard theorem, conditional application

If D is a genuine [N,K]{16} linear code with D^(perp_H) D, the usual Hermitian construction over F{16}= F_(4) gives a 4-ary stabilizer code of dimension K_q=2K-N. Its quantum distance is the minimum weight of a nontrivial logical coset (commonly represented by Dsetminus D^(perp_H)); it is at least the minimum distance of D, but equality is not automatic. The quantum Singleton bound K_q+2D_q€ N+2 and defect N+2-K_q-2D_q are standard once actual parameters are known. No such D or parameters have been established here.

11. New lengths unsupported

For a fixed map to F ^{4kn}, the image length is N=4kn, independent of . A different multiplier may alter code existence or dimensions only after a valid construction is defined. The claim that lengths are reachable only for ne1 is not established by the blueprint and conflicts with its own length formula if it means the resulting quantum block length.

Bottom line

The only unconditional numerical identity in the blueprint's main algebraic chain is the cardinality/symbol-count identity. Field-level separability, the fifth-root count, the = identity, and the quantum Singleton formula are conditionally correct. Their application to M (A) depends on foundational structures the blueprint has not defined and, in the local-ring case, contradicts the actual matrix-ring quotient.

Algebraic framework audit

1. The ring actually named in the blueprint

Set A = F [u]/ u^k and R =M (A). Its basic structure is:

|A |=2^k, and |R |=(2^k)^{16}=2^(16k)=16^(4k).
Z(R)=A I , the scalar matrices.
J(R)=M (uA); for k=1, this is zero.
R /J(R) • M (F), a simple noncommutative matrix ring, not a field.
R is noncommutative and not local. A matrix is a unit exactly when its reduction in M (F) is invertible.

The claim that R is the commutative local ring F [u,v]/(u^k,v) is impossible: the latter has residue field F , whereas the matrix ring has quotient M (F) by its Jacobson radical. Equal cardinality does not imply isomorphism.

2. What the F -coordinate description can mean

A copy of F can be embedded in M (F), and the matrix ring can be expressed using cyclic-algebra/module coordinates. Such coordinates are useful but are not central scalars. The 2026 preprint's cyclic-algebra discussion includes ae=e (a) and e =1; setting v=1+e gives a nilpotent coordinate, not a central polynomial variable commuting with all of F . Thus

F +v F +v F +v, F

may describe an additive normal form/module, but the blueprint cannot replace the matrix ring by the ordinary commutative ring F [v]/(v). The separate scalar u commuting with the matrix entries does not make v central.

3. Central versus noncentral constamultipliers

A central unit of R has the form aI with a*f* A ^ . For k=1, A ^ ={1}; for k>1, |A ^ |=2^(k-1). There is no central order-5 unit. The four nontrivial fifth roots in an embedded F are noncentral matrix units.

This is not a claim that no noncentral constacyclic shift can be defined. For right-R -linear codes, the coordinate shift with **left** multiplication by a fixed is right-linear. But the blueprint must explicitly choose this convention and prove the desired properties. With the ordinary central-indeterminate polynomial ring R [x], the element x - is not central for noncentral ; its generated ideal need not be two-sided. Consequently, the usual ideal correspondence and the displayed quotient-ring/CRT argument do not follow. A skew-polynomial, endomorphism, or direct-module formulation would be a different framework requiring approval and proof.

4. Why the proposed component rings do not follow

The blueprint moves from factorization over F to

S =K [u,v]/ u^k,v , K = F [x]/ f .

These are commutative local complete intersections. Their cardinalities match the sizes one might expect for some module components, but the quotient R /J(R) and the noncentral cyclic-algebra action rule out the claimed ring isomorphism. A module decomposition, if available, has to retain the left/right action and the noncommuting field coordinates; cardinality alone is insufficient.

The following proposed steps are therefore blocked as statements about R :

1. Hensel-lifting a factorization in a residue **field** F to the noncommutative matrix ring as if it were a commutative local algebra.
2. Identifying every code component with an ideal of S .
3. Classifying all codes by ideals in S and multiplying component cardinalities.
4. Applying commutative monomial-ideal annihilator formulas directly to matrix-ring codes.

A valid replacement might use right-module decompositions rather than ring CRT components, but this audit does not construct or prove one.

5. Duality requires a precise module/form convention

The blueprint needs to specify all of the following before its Euclidean or Hermitian dual formulas can be audited as theorems:

whether codewords are left or right R -modules;
whether the constacyclic shift acts by left or right multiplication and on which coordinate;
the value ring/field of the inner product and multiplication order;
the map called conjugation, including whether it is a ring automorphism or an involutive anti-automorphism;
whether the dual is a left or right submodule and which annihilator is meant;
how the chosen form behaves under the proposed Gray map.

The rule a a is a field automorphism of F with fixed field F . It is not yet a definition on all of R . For a noncommutative Hermitian module form, the extension must respect the skew multiplication and the chosen module side. It is not enough to say that u,v are fixed.

6. What remains mathematically sound

The size |R |=16^(4k) and the resulting cardinality-compatible field-symbol count are correct.
The field F can be used as a coordinate field after a specific module action and map are fixed.
The commutative ring K[u,v]/(u^k,v), if it were the chosen algebra, is a finite local Frobenius ring; monomial ideals there can be counted by their monomial basis.
The standard field Hermitian construction can be used after an actual F -linear image and dual-containment statement are proved.

These observations do not rescue the present chain of arguments. **A researcher must first approve a corrected algebraic framework; no alternative construction is implemented here.**

Source checks

The primary structural source inspected was Joshi Patel Prakash, arXiv:2602.18255v1, particularly its cyclic-algebra relation, ring/module description, and Sections 3 5. It is a preprint, not proof that the blueprint's commutative interpretation is valid. The 2022 M article's publisher abstract and correction were also checked: article, correction.

Literature and prior-art audit

Purpose: assess the blueprint's cited gap, not certify an exhaustive literature search. Search and source review were current to 12 November 2026. No quantum-code parameter search was run.

1. Direct matrix-ring prior art

M (F +u F): the k=2 case

Patel, Prakash and Islam, Cyclic codes over M (F +u F), Cryptography and Communications 14 (2022), 1021 1034, DOI 10.1007/s12095-022-00572-9. The publisher abstract states that this exact ring is studied, with cyclic-code structure, Euclidean and Hermitian duals, and a Gray image whose codes are F -linear. This is the k=2 member of the blueprint's matrix-ring family and is material prior art for its claimed ring/cyclic/duality/Gray-map foundation.

A published correction appeared 24 May 2022: DOI 10.1007/s12095-022-00584-5. The correction lists twelve equation/display corrections, including generator expressions and the displayed Gray-map matrix. Any reuse or comparison must consult the corrected version. The correction notice describes presentation/formula-display errors; this audit does not infer from that notice that every underlying theorem has been independently revalidated.

Access limitation: the full 2022 article body was not available in the workspace for line-by-line theorem verification. Its publisher metadata/abstract and the complete correction notice were reviewed. The existence and stated scope of the prior work are nevertheless clear.

M (F +u F +u F): the k=3 case

Joshi, Patel and Prakash, Construction of new codes over F from cyclic codes over M (F +u F +u F), IEEE International Symposium on Information Theory (ISIT 2025). This conference paper is cited as reference 19 in the 2026 Quantum Information Processing article. It directly covers the k=3 cyclic-code specialization. The ISIT 2025 proceedings table of contents lists the paper; see proceedings listing.

General k, cyclic codes over M

Joshi, Patel and Prakash, Construction of Cyclic Codes over a Class of Matrix Rings, arXiv:2602.18255v1, submitted 20 February 2026. The preprint treats M (F [u]/ u^k), cyclic codes (odd-length structural results), Euclidean/Hermitian dual formulas, and Bachoc/Gray maps to F . It is direct overlap for the blueprint's cyclic baseline across general k, though it does not by itself settle the proposed constacyclic extension or its quantum application. No journal version was confirmed in the sources checked; cite it as a preprint unless that status is rechecked.

The attached arXiv preprint's Theorem 4.1 is specifically an **odd-length cyclic Euclidean-dual** statement. Its proof presents a candidate dual submodule, establishes an inclusion by generator orthogonality, then compares cardinalities. It is not a theorem for nontrivial constacyclic multipliers and does not prove the blueprint's new component/Gray/quantum claims.

2. Matrix-ring quantum precedent

Joshi, Patel and Prakash, Quantum codes construction from cyclic codes over matrix rings, Quantum Information Processing 25, article 207 (2026), DOI 10.1007/s11128-026-05230-0, published 9 June 2026. It studies cyclic codes over M (F_p[u]/ u^k), and for a matrix ring over F gives a CSS route using a new non-degenerate inner product and a suitable Gray map. This is not the same alphabet or constacyclic family as the blueprint, but it is close matrix-ring/quantum prior art. Its full text expressly notes that an earlier Gray map did not transfer the required dual-containment property and introduces a different map/product pair. That makes the blueprint's Gray-map orthogonality claim a substantive proof obligation.

The 2026 article also cites both the 2022 M cyclic-code paper and the 2025 k=3 ISIT paper. Those works must appear in any current account of the gap.

3. Constacyclic/quantum adjacent work

These works show that constacyclic codes over finite non-chain rings, Hermitian dual containment, Gray maps, and quantum constructions are established research themes. They are not direct duplicates of M (F [u]/(u^k)), but they make a novelty claim based only on combining the words constacyclic and quantum untenable.

Alahmadi et al., New quantum codes from constacyclic codes over a non-chain ring, Quantum Information Processing 20, article 60 (2021), DOI 10.1007/s11128-020-02977-y.

Islam, Patel and Prakash, A family of constacyclic codes over a class of non-chain rings A_(q,r) and new quantum codes, Journal of Applied Mathematics and Computing 68 (2022), 2493 2514, DOI 10.1007/s12190-021-01623-9. The article studies constacyclic and skew constacyclic codes and Hermitian-dual-containing conditions over a commutative non-chain family.

Yadav et al., Hermitian hull of constacyclic codes over a class of non-chain rings and new quantum codes, Computational and Applied Mathematics 43, article 269 (2024), DOI 10.1007/s40314-024-02789-1.

Tian, Gao and Gao, Hulls of constacyclic codes over finite non-chain rings and their applications in quantum codes construction, Quantum Information Processing 23, article 9 (2024), DOI 10.1007/s11128-023-04230-8.

Tang et al., Quantum codes from constacyclic codes over finite non-chain rings, Quantum Information Processing 25, article 305 (2026), published 3 September 2026, DOI 10.1007/s11128-026-05334-7. Its ring is a commutative finite non-chain family, not the blueprint's matrix ring; it is current adjacent literature.

4. Earlier matrix cyclic-code context

The 2026 preprint cites earlier work including Pal, Bhowmick and Bagchi, Cyclic codes over M (F), Journal of Applied Mathematics and Computing 60 (2019), 749 756, DOI 10.1007/s12190-018-01235-w, as well as work on M matrix rings. This is relevant background for the matrix-ring/ F coordinate viewpoint, not evidence that the proposed quantum extension is either novel or non-novel.

5. Audit of the blueprint's literature statement

The blueprint says the matrix-ring paper covers only cyclic codes and the QIP paper covers M quantum codes. Those scope descriptions are broadly accurate for the cited papers, but the gap argument is incomplete because it omits the exact k=2 published article's correction, the k=3 ISIT 2025 paper, several general constacyclic/quantum precedents, and current 2026 work. The combination was not found in either introduction is not a novelty test.

A targeted search did **not** establish a prior article exactly matching M , noncentral order-5 constamultiplier, Hermitian Gray image, and F quantum codes. It also did not establish that no such work exists. The search was not exhaustive, and the blueprint's multiplier/algebra framework is not yet coherent enough to define a stable novelty query.

6. Literature-review limits and required standard

Search date: 12 November 2026; web metadata/abstracts and the supplied PDFs were reviewed.

The 2026 arXiv preprint and QIP article were read in the supplied/extracted versions for relevant sections; the 2022 full article body was not available for complete theorem-by-theorem review.

No citation chaining, database-by-database systematic review, or code-table search was performed.

Before claiming novelty, a researcher must compare exact ring, module side, multiplier class, length hypotheses, dual form, map, and quantum parameters against the direct and adjacent works above, and document the search protocol. Novelty cannot be certified from this audit.

Novelty and research-gap audit

Executive assessment

The blueprint's novelty claim is not established. The exact proposed conjunction „ an appropriately defined constacyclic family over the stated M -matrix ring, a valid field image, and quantum codes „ was not identified as an exact duplicate in the targeted search. That is not proof of novelty. The claimed gap omits direct prior work on the same matrix-ring family and has a foundational mismatch: the order-5 multipliers are not central scalars of the stated ring, while the proposed local-ring quotient model is not the matrix ring.

Claim-by-claim novelty audit

The matrix-ring paper studies only cyclic codes and stops at classical F codes.

This is a fair description of the 2026 arXiv v1 preprint's stated scope: cyclic codes, dual formulas, and Gray/Bachoc images over F . It does not mean the cyclic/dual/Gray foundations are new. The 2022 k=2 article and 2025 k=3 ISIT paper cover direct specializations; the 2026 preprint covers general k as a preprint.

The quantum paper does CSS only for M .

The 2026 Quantum Information Processing article's matrix-ring setting is M , not M , and it uses a CSS route with a new inner product/map. That is adjacent, not a direct duplicate. It is nevertheless close prior art for quantum construction from cyclic matrix-ring codes and demonstrates that dual transfer through a Gray map is not automatic.

M + constacyclic + quantum is not covered in either introduction.

This phrase is not enough to establish a research gap. A novelty review must check full texts, references, current publications/preprints, and adjacent formulations, not only introductions. The 2025 ISIT k=3 paper and 2022 corrected k=2 article are not mentioned in the blueprint's stated gap. The 2021 2026 constacyclic/non-chain-ring quantum literature also covers the general methods and expected claims.

Replacing cyclic by constacyclic opens new lengths and codes cyclic codes cannot reach.

No theorem or example supports this. The claimed image length N=4kn is independent of ; the same k,n gives the same block length. A different shift might yield a different code family in a valid setup, but new lengths and superiority to every cyclic family require a precise comparison and evidence. Moreover, the proposed fifth-root multipliers are noncentral in M (A); the blueprint's polynomial-ideal route does not accommodate them as written.

Prior-art map

Same ring, cyclic codes, duals and F -linear Gray images: Patel Prakash Islam (2022), exact k=2, with a published correction.

Same ring family, cyclic codes: Joshi Patel Prakash (ISIT 2025), exact k=3 case.

Same ring family, general k, cyclic codes and dual formulas: Joshi Patel Prakash (arXiv:2602.18255v1, 2026 preprint).

Matrix-ring quantum construction: Joshi Patel Prakash (QIP 2026), M rather than M , with CSS and a map designed to transfer a nonstandard dual.

Constacyclic/Hermitian/quantum methods on finite non-chain rings: papers from 2021, 2022, 2024, and September 2026 listed in the literature audit. These are adjacent families, not the same matrix ring.

What might remain as a research gap

A carefully defined M constacyclic theory using an admissible multiplier class and a proved map/inner-product bridge might contain new results. This audit does **not** certify that gap. The blueprint's current candidate family does not yet provide a valid object on which to assess novelty: it conflates a matrix ring with a commutative local ring, and it does not specify the right/left action for noncentral multipliers. Novelty must be reassessed only after those choices are approved and the claimed theorem statements are made precise.

Minimum standard for a later novelty claim

1. State the exact ring presentation with all cross-relations and identify central versus noncentral elements.
2. Define the code module side, multiplier class, shift, and polynomial/endomorphism model.
3. State a theorem that is genuinely new relative to the cyclic results at k=2,3 and general k, and separate structural novelty from new parameter tuples.
4. Compare the exact duality, Gray map, and quantum construction against the 2026 M CSS method and established constacyclic quantum work.
5. For each claimed new quantum code, compare exact (N,K_q,D_q) parameters against current sources, using exact distances or clearly labelled certified bounds.
6. Record database/search terms, dates, citations, and excluded near-matches. Do not treat a novelty search as a theorem proof.

Decision

Novelty: unverified, not presently defensible as claimed. This is not a finding that no novel contribution can exist; it is a finding that the supplied blueprint has not demonstrated one. See `04_LITERATURE_AND_PRIOR_ART_AUDIT.md` for source details and limitations.

Methodology and feasibility audit

Overall assessment

The blueprint identifies reasonable work categories „ structure, cardinality, duality, a Gray map, and a later parameter study „ but the dependency order is unsound. The proposed computations assume a commutative local component model that is not the specified matrix ring. Consequently, the search pipeline cannot currently validate its algebraic inputs. This document assesses the plan only; no research execution or parameter/code search was performed.

Dependency audit

A. Factorization and component decomposition

The blueprint factors x - over F , invokes Hensel lifting, and writes a CRT sum of commutative rings K[u,v]/(u^k,v). This requires the correct coefficient ring, a valid central polynomial relation, and the appropriate factorization/lifting theorem. The actual M (A) has radical quotient M (F), not F . A module decomposition may be possible in a corrected framework, but the displayed commutative CRT argument is not established.

Feasibility gate: stop until the ring model, side conventions, and decomposition have been fixed and proved or explicitly restricted to a valid subproblem.

B. Ideal classification and cardinality

Monomial ideals in the proposed commutative truncated polynomial ring are finite and countable by monomial basis elements. This is a feasible restricted family **if** that ring is genuinely the component algebra. It cannot be advertised as classification of all constacyclic codes or used to prove a general necessary-and-sufficient condition. A Gr…bner-basis plan would need a specified coefficient field, monomial order, quotient ideal, reduction rules, and a proof that the computed normal forms cover every desired component. No such implementation was attempted.

C. Duality and component pairing

The component map, conjugate reciprocal, annihilator direction, and root pairing depend on the valid algebra and inner product. A single field-level rule is not enough for a noncommutative right module. The blueprint needs to define left/right annihilators and show that the chosen dual operation preserves or permutes the actual components. The dual-containing test must be proved before it can drive enumeration.

D. Gray map and metric

The plan correctly recognizes that map definitions must be read before reuse. It nevertheless treats four distinct properties as if they were a single Gray map result:

bijectivity/cardinality;
additivity and F -linearity;
Hamming-distance preservation for the chosen ring weight;
preservation of the exact dual/orthogonality relation required by the quantum construction.

Each needs a separate statement and proof. The quasi-twisted index also requires an intertwining calculation for the shift. The 2026 M quantum paper uses a special inner product and a separate Gray map precisely because its earlier map does not transfer the required dual.

E. Quantum parameters and distance

If the field image is a valid [N,K]_{16} code and is Hermitian dual-containing, the standard construction gives quantum dimension 2K-N. The quantum distance is not automatically the exact classical minimum distance: it is determined by nontrivial logical cosets and is at least the classical minimum distance in the usual setup. The blueprint should report exact distance only when established; otherwise report a certified lower bound and use consistent notation. The Singleton defect formula itself is standard.

F. Proposed computation grid

The stated range k *f* {2,3,4}, odd nle25, gives N=4knle400. This arithmetic is fine, but the grid overlaps direct prior work at k=2 and k=3, and its fifth-root multipliers cannot be treated as central. Distance computation can be expensive at those lengths; no runtime/equivalence/certification plan is supplied. A database comparison must include a timestamp, exact versus lower-bound status, quantum dimension convention, and a reproducible record of each generator and map.

Constraint observed: no codeword enumeration, Magma/GAP/Python run, quantum-parameter search, or code-table search was performed.

Literature and publication feasibility

The blueprint's references and gap discussion omit direct k=3 ISIT 2025 work, the corrected status of the 2022 k=2 paper, and current 2026 quantum matrix-ring literature. Target journal names are not a substitute for a completed novelty audit. Q1 quartiles depend on database/category/year and were not certified here. In its present state, the blueprint is not ready for a Q1 submission or even research execution.

Feasible order after explicit approval

1 Decide and state the actual algebraic object, its center/radical, and the F -action.
2 Define left/right code modules and a constacyclic shift that is compatible with that choice; distinguish central and noncentral multipliers.
3 Choose a valid polynomial, skew-polynomial, or endomorphism framework. Do not assume ordinary quotient ideals for noncentral .
4 Prove a restricted structural/cardinality result in that framework. Clearly label monomial or other subclasses as partial.
5 Define the dual form/involution and prove the dual operation for the same shift.
6 Define the exact field map and verify linearity, distance, shift, and dual transfer separately.
7 Apply the appropriate field quantum construction and define exact distance versus lower bound.
8 Re-run the literature review and novelty comparison before any code search.
9 Only then, if approved, conduct a reproducible search and compare exact parameters.

This is a recommendation roadmap, not execution or permission to modify the blueprint.

Theorem verification audit

Scope: verify the blueprint's proposed theorem statements against the ring structure and the supplied/cited source material. No new theorem is developed or proved here. Conditional means a standard statement may hold after its missing hypotheses are supplied; it is not verified for the blueprint's matrix ring as written.

A. Proposed ring and unit statements

Claim A1 M (F [u]/(u^k)) is a local ring with residue F

Verdict: false. For A = F [u]/(u^k), 𝕁(M (A))=M (uA), and the quotient is M (F), not a field. The center is A_kI . The commutative local ring proposed in the blueprint cannot be isomorphic to the matrix ring.

Claim A2 unit iff the F -residue is nonzero

Verdict: false for the stated ring. The correct matrix-ring criterion is invertibility after reduction modulo u in M (F). The blueprint's criterion would require the false local-ring model.

Claim A3 four order-5 elements are permissible central 's

Verdict: field count correct; central-ring use false. F ^{ } has four elements of order 5, but an embedded F is noncentral. The central-unit group comes from A ^ , of order 1 for k=1 and 2^(k-1) for k>1; it has no element of order 5.

B. Proposed structure/cardinality theorems

Claim B1 odd length gives a separable factorization and a CRT decomposition into S =K [u,v]/(u^k,v)

Verdict: field separability is conditionally correct; the ring decomposition is unsupported and conflicts with the matrix ring. Odd n makes x - separable over a characteristic-2 field when ne0. It does not turn M (A) into a commutative local algebra or justify the displayed factor rings. For noncentral , the ordinary polynomial-quotient setup also needs a separate definition.

Claim B2 classify ideals by monomial/principal ideals and count by monomial bases

Verdict: conditional for the proposed commutative quotient, not verified for R . In a justified component K [u,v]/(u^k,v), monomial ideals can be counted by monomial basis elements. This only treats that subfamily. A Gr…bner-basis suggestion is a research method, not a proof of completeness or feasibility. The ring-component premise has not been established.

Claim B3 |C|=prod |K |^{m }

Verdict: conditional. It follows from an actual direct-product component decomposition and the stated component sizes. Neither condition has been proved for the matrix-ring codes as formulated.

C. Proposed duality and dual-containment theorems

Claim C1 Euclidean dual is -constacyclic; Hermitian dual is -constacyclic

Verdict: conditional field-level result, not verified for the matrix ring. Under the usual field shift/form convention, the Hermitian multiplier is ^{-1}= . If =1, then = . For the noncommutative ring one must define the module side, multiplier action, involution/anti-involution, form, and product order first.

Claim C2 components pair under a shifted root map j -4j

Verdict: unproved and underspecified. It requires an explicit root parametrization, a valid factorization model for x - , and an induced conjugate-reciprocal action on the actual components.

Claim C3 dual containment iff overline{Ann(C_(i'))} C for all i

Verdict: target only. No proof is supplied. Ann must be designated left or right, and the dual/component identification must be established. The result cannot be inferred from a commutative model that is not the stated ring.

D. Gray-map and quantum theorems

Claim D1 there is an F -linear distance-preserving :R F ^{4kn}

Verdict: cardinality is compatible; the map properties are unverified. A specific F -action, map, metric, and shift action are required. Isometry, additivity, linearity, quasi-twisted structure, and dual transfer must be checked separately.

Claim D2 (C^(perp_H))= (C)^{perp_H}

Verdict: unproved. Requires the full definition of , the ring form, the field Hermitian product, and a compatibility proof plus appropriate size/dimension facts. The cited QIP M work uses a specially selected inner product and a different Gray map because its earlier map did not transfer duals.

Claim D3 Hermitian construction gives [[4kn,2K-4kn, d_H((C))]]_4

Verdict: standard conditional dimension/lower-bound statement. It applies only if D= (C) is genuinely F -linear and Hermitian dual-containing. The exact quantum distance is determined by nontrivial logical cosets and may exceed the classical distance; equality/purity must be shown. No eligible D is established in the blueprint.

Claim D4 quantum Singleton defect =N+2-K_q-2D_q

Verdict: correct definition under the usual stabilizer-code convention. The formula is not evidence for the existence or novelty of any parameters.

E. Source-paper theorem check

The attached arXiv:2602.18255v1 states Theorem 4.1 for **odd-length cyclic** codes over its matrix-ring model, gives a Euclidean-dual expression, and proves it by a candidate inclusion followed by cardinality comparison. The inspected text supports that scope and proof pattern; extracted/rendered formulas and index conventions must be read carefully. It does not state the blueprint's nontrivial -constacyclic theorem, its shifted component criterion, or its F quantum result. The source paper is a preprint and cannot be cited as verification of those extensions.

The 2022 M published correction (DOI) corrects displayed equations and a Gray-map matrix. It does not supply the blueprint's constacyclic/quantum theorem.

QIP 2026 source claims sampled in the supplied PDF

The extracted source defines a reversed-index product a,brangle_R=sum†‡^{n-1}a b , described there as nondegenerate. In the sampled sections, Theorem 12 states a dual-size formula; Theorem 13 gives a dual-containing criterion involving self-reciprocal factors and selected ideals; Theorem 14 claims distance-preserving Gray maps phi ,phi for the source's k=3,2 cases, respectively; and Theorem 15 claims ˆ(C_R^‰)=ˆ(C)_R^‰. These are claims in the M paper with its specified product and maps, not results for the blueprint's M setting.

The extracted Theorem 15 proof begins with an orthogonality/product condition (the extracted line reads ab=0) and expands coordinates, but the available extract ends mid-argument. The product order, side conventions, and complete proof were not independently certified from the rendered source pages here. Therefore the theorem statements are recorded as adjacent-source claims only; this audit does not endorse their proof or transfer them to the blueprint. The reversed-index product must not be silently replaced by the standard coordinatewise Euclidean/Hermitian product.

Verification conclusion

No proposed theorem that depends on the commutative local-ring model is verified as a theorem about M (A). The standard field and quantum statements remain usable only after the researcher supplies and proves the missing algebra, shift, form, and map hypotheses. This audit stops short of constructing those hypotheses or proving replacements.

Error and risk register

Severity key: **P0 = foundational stop issue**; **P1 = major theorem/novelty blocker**; **P2 = execution, reproducibility, or publication risk**. Required action means a recommendation for a future researcher-approved revision, not work performed in this audit.

ID	Severity	Blueprint claim / location	Finding and consequence	Required action after approval
E01	P0	pp. 1 2: R is local with residue F , maximal ideal u,v	**False.** J(M (A))=M (uA); quotient is M (F). Invalidates the proposed residue-field, Hensel, local-component, and unit arguments.	Re-state ring, center, radical, quotient, and exact cyclic-algebra/module presentation.
E02	P0	p. 1: unit iff nonzero F -residue	**False for the matrix ring.** Unit status is determined by invertibility of the mod-u matrix in M (F).	Replace with the correct matrix-ring criterion in the approved setup.
E03	P0	pp. 1 3, 5: order-5 's used as scalar constamultipliers	The order-5 elements of an embedded F are noncentral. The center's unit group has no order-5 element. Ordinary quotient/ideal machinery for noncentral is not defined.	Choose central multipliers or explicitly define a noncentral shift and a compatible skew/module framework.
E04	P0	pp. 2 3: commutative S =K [u,v]/(u^k,v) CRT components	No ring isomorphism from the stated matrix ring has been shown; the proposed local component ring has a field residue, unlike the matrix-ring quotient.	Rebuild component decomposition with correct side/action; do not infer from cardinality.
E05	P0	pp. 2 4: Hermitian conjugation on all of R	The field automorphism a a is not a complete definition of an involution on the noncommutative matrix ring or of a Hermitian module form.	Specify and verify the full-ring involution/anti-involution, module side, form, and output field pairing.
E06	P0	p. 4: (C^(perp_H))= (C)^{ perp_H}	No map/form compatibility proof. Without it, a ring dual-containing code need not give the field dual-containing code required by the quantum construction.	Define the map and prove linearity, metric, and exact dual-transfer properties separately.
E07	P1	pp. 2 3: Hensel/CRT classification and all-ideal theorem	The steps rely on the wrong residue/component ring and do not resolve noncentral multiplier quotient issues.	Re-derive only after the algebraic framework passes a stop gate; state scope if partial.
E08	P1	p. 2: monomial ideals followed by principal/all ideals	Monomial enumeration is a restricted subfamily. It cannot prove a classification or global necessary-and-sufficient criterion. The Gr…bner-basis sentence is a plan, not a theorem.	Separate a proved subclass theorem from any all-ideal claim; give completeness proof.
E09	P1	p. 3: component map j -4j, annihilator criterion	Root parametrization, component action, left/right annihilator, and proof are missing.	Define component indices and annihilator side; prove the dual/component correspondence.
E10	P1	p. 4: F -linearity, quasi-twisted index, distance preservation	A size-compatible or weight-isometric Gray map does not imply these properties.	State the scalar action and prove each map property, including shift intertwining.
E11	P1	pp. 1, 5: constacyclic gives new lengths inaccessible to cyclic codes	Unsupported. The formula N=4kn is independent of ; no comparison or example is supplied.	Rephrase as a testable hypothesis and compare actual valid parameter families.
E12	P1	pp. 1, 6: novelty gap stated from two introductions	Misses direct k=2 corrected article and k=3 ISIT 2025 paper; current M quantum and broad constacyclic work is adjacent prior art.	Complete a current, systematic literature review after the object is defined.
E13	P1	p. 4: exact quantum distance from d_H((C))	Standard construction gives a distance controlled by nontrivial logical cosets; the classical distance is a lower bound, not automatically exact.	Define the quotient distance; report certified bounds unless exactness is proved.
E14	P2	pp. 5 6: Magma/GAP/Python and length range	Feasibility, exact-distance certification, runtime, and reproducibility are unspecified. Computation cannot validate the algebraic premises.	Defer all search until proofs/maps are validated; document tools and exactness.
E15	P2	p. 5: Grassl status new/matches/dominated	Comparison rules and snapshot date are not fixed; new requires exact parameter and code-table criteria.	Record database/version/date, exact versus lower-bound distances, and dominance rule.
E16	P2	p. 7: candidate Q1 venues	Quartile varies by source/category/year; journal list does not establish readiness.	Verify current category-specific rankings only after a valid theorem and novelty case exist.

ID	Severity	Blueprint claim / location	Finding and consequence	Required action after approval
E17	P2	2022 source cited without correction	The publisher lists a correction with twelve equation/display corrections, including a Gray-map matrix.	Cite and use the corrected record; do not reuse the uncorrected display blindly.

Critical contradiction log

1 **Locality vs matrix quotient:** the blueprint says residue field F ; the actual radical quotient is M (F). These statements cannot both hold for the same ring.

2 **Central order-5 scalar vs matrix center:** fifth roots exist in a noncentral embedded field; central units of R have 2-power order. The proposed order-5 scalar family cannot be central in the stated ring.

3 **Commutative CRT components vs skew cyclic-algebra model:** the field-coordinate normal form does not make the ring commutative; the stated local component rings are not established.

4 **Ring dual vs field Hermitian dual:** a Gray map's cardinality/weight properties do not imply the exact orthogonality transfer required by the quantum theorem.

5 **New lengths vs fixed image length:** N=4kn does not depend on .

Overall risk rating

Critical / no-go for current execution. E01 E06 block the central mathematical chain. No computational search or blueprint modification was performed.

Recommended correction plan

Status: recommendations only. The audit is complete; no correction, construction, proof, or computation has been carried out. Stop here and wait for explicit researcher approval before any follow-up work.

Phase 0 freeze the current blueprint

Do not use the current F [u,v]/(u^k,v) local-ring model, order-5 scalar family, component cardinality formulas, Gray-map duality claim, or quantum parameters as established results. Preserve the uploaded PDF unchanged. Mark the current statements as rejected/conditional in any future draft.

Gate 0: researcher approves a revised algebraic direction before work resumes.

Phase 1 settle the algebra before coding theory

1. Fix the object R=M (F [u]/(u^k)) and document its center, radical, quotient, units, and cyclic-algebra presentation, including all cross-relations.
2. State whether codes are left or right R-submodules.
3. Define the exact constacyclic shift, multiplier side, and what -constacyclic means for central and noncentral .
4. If the multiplier is noncentral, do not use an ordinary central-variable quotient as if it were a ring. Choose and justify a compatible module, skew-polynomial, or shift-operator model.
5. Revisit the proposed order-5 values. If centrality is required, remove that family for this ring; if noncentral shifts are intentionally used, state that explicitly and rework every later theorem accordingly.

Gate 1: a coherent object/shift/module definition and correct ring facts.

Phase 2 establish structure and cardinality in the corrected setting

1. Prove the exact decomposition used; do not replace a module decomposition with a ring CRT isomorphism without proof.
2. State the classification scope precisely. If only monomial ideals or a selected class is treated, say so in the theorem title and novelty claim.
3. Derive code sizes from the actual component/module structure and verify no components are double-counted.
4. Treat Hensel lifting only after identifying a valid commutative central coefficient subring or a proved noncommutative analogue.

Gate 2: independently checked structure and cardinality statements for the chosen scope.

Phase 3 define duality and prove the exact dual statement

1. Specify the value of the form, multiplication order, module side, and involution/anti-involution on the whole ring.
2. Define Euclidean and Hermitian duals distinctly; name left/right annihilators.
3. Derive the dual shift parameter from the chosen convention rather than importing or without checking.
4. Establish any reciprocal/conjugate component permutation and prove the dual-containment criterion.

Gate 3: a precise dual theorem and a proved dual-containing condition.

Phase 4 validate the field-image route

1. Give the exact map and the F -action on its domain.
2. Prove bijectivity/cardinality, additivity/field-linearity, and metric preservation as separate properties.
3. Compute the induced image of the constacyclic shift; state the quasi-twisted structure only if the block action proves it.

4 Prove the exact dual-transfer identity (or a sufficient inclusion plus verified cardinality relation) for the chosen field Hermitian product. If this cannot be done, do not claim the Hermitian quantum route.

Gate 4: a genuine F -linear dual-containing image or a different approved quantum framework.

Phase 5 quantum statements and literature positioning

1 State quantum dimension as 2K-N only after a valid [N,K]_{16} image and Hermitian dual containment are established.

2 Define quantum distance via nontrivial logical cosets. Label a certified lower bound separately from an exact distance; prove any pure equality claim.

3 Update the prior-art comparison to include the corrected 2022 k=2 paper, 2025 ISIT k=3 paper, general-k 2026 preprint, 2026 M quantum paper, and current constacyclic/non-chain quantum literature.

4 Replace new lengths cyclic cannot reach with a precise, theorem-supported comparison or remove it.

Gate 5: a defensible contribution statement and novelty claim, reviewed by the researcher.

Phase 6 computation and submission readiness (only if later approved)

Only after Gates 0 5 pass should a researcher approve any Magma/GAP/Python enumeration or code-table comparison. The search must specify field encoding, generator convention, exact/lower-bound distance status, the database snapshot, dominance rule, and reproducibility data. Q1 venue selection and quartile claims come last and must be checked by year/category/source.

Stop conditions

Stop and return to the researcher if the desired -family is required to be central but no nontrivial order-5 central multiplier exists; if the intended Gray map fails the needed dual transfer; or if the corrected ring cannot support the claimed component decomposition. These are research-direction decisions, not automatic invitations to change the user's project.

Final audit decision

Decision

REJECT AS SUBMITTED DO NOT PROCEED WITH THIS BLUEPRINT IN ITS CURRENT FORM

This is the audit's no-go decision for the submitted blueprint, not a conclusion that all research on constacyclic codes or quantum codes over matrix rings is impossible. A substantially revised proposal may be considered only after explicit researcher approval.

Direct answers

Is the blueprint reliable?

No. Its key ring claim is wrong: M (F [u]/(u^k)) is noncommutative and not local, with radical quotient M (F), not residue field F . The proposed unit test, order-5 central multipliers, local CRT factors, and downstream ideal/cardinality framework therefore do not describe the named matrix ring. Duality and Gray-map transfer are also left as goals rather than proved properties.

Is it novel?

Not established. The k=2 cyclic-code case has a 2022 journal article and correction; the k=3 case has an ISIT 2025 paper; and a 2026 preprint treats cyclic codes for general k. A 2026 QIP paper studies quantum construction from cyclic codes over M -matrix rings, alongside a substantial existing constacyclic/non-chain-ring quantum literature. These do not prove that a valid M constacyclic/quantum theorem cannot be novel, but the blueprint has not demonstrated a distinct, valid contribution.

Is it ready for research execution or a Q1 submission?

No. The algebra, multiplier convention, polynomial/module model, involution and inner product, dual theorem, Gray-map properties, novelty position, and exact-distance methodology must be resolved first. Q1 quartile status is also database/category/year-dependent and does not remedy missing mathematical foundations.

Basis for the no-go decision

1. A foundational matrix-ring/local-ring contradiction affects every later section.
2. The chosen order-5 multipliers are not central units of the stated ring, so the proposed ordinary polynomial-ideal framework is not justified.
3. The structural, cardinality, dual-containment, and Gray-map theorems are mostly targets contingent on that invalid framework.
4. The quantum construction is standard only after a valid F -linear Hermitian dual-containing image exists; the blueprint does not establish one.
5. Prior art is material and incompletely represented; novelty remains unverified.

Scope and stop point

The requested audit files and consolidated PDF/ZIP are the final deliverables. No paper was written, no construction or theorem was developed, no computation or code-parameter search was run, and no correction was applied to the uploaded blueprint. The original uploaded PDF remains unchanged. **Stop here and wait for explicit researcher approval** before any literature expansion, blueprint modification, proof work, implementation, or parameter search.

Primary references

Joshi, Patel & Prakash, arXiv:2602.18255v1, cyclic codes over a class of matrix rings, submitted 20 February 2026.

Patel, Prakash & Islam, Cyclic codes over M (F +u F) , 2022, with published correction.

Joshi, Patel & Prakash, Quantum codes construction from cyclic codes over matrix rings , Quantum Information Processing 25, 207 (2026).
Joshi, Patel & Prakash, Construction of new codes over F from cyclic codes over M (F +u F +u F), IEEE ISIT 2025; cited in the 2026 QIP paper, Ref. 19.